

Building a Framework for Comprehensive CXM: Opportunities and Challenges for Web-based Services

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Abstract

Customer Experience Management (CXM) focuses on delivering consistent and high-quality experiences across customer touchpoints, aiming to cultivate loyalty and drive long-term business success. Effective CXM requires analyzing customer behaviors and perceptions. However, achieving holistic management of customer experiences remains challenging, even with advances in data analytics and digital transformation. Through a literature review, we developed a metrics framework to measure and manage customer experiences in web-based services comprehensively. This framework addresses key dimensions such as subjects, phases, and aspects of experience. Practitioners generally provided positive feedback on the framework's appropriateness and shared practical insights for real-world services. They emphasized the need for flexible adaptation of metrics and the importance of presenting data effectively. Additionally, we envision an AI-Enhanced Collaborative CXM model that integrates AI's analytical capabilities with practitioners' domain knowledge to create tailored insights. This model offers a promising direction for enhancing CXM strategies through a more collaborative, data-driven approach.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Customer Experience Management, Metrics Framework, Web-based Services

ACM Reference Format:

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1 Introduction

Customer Experience Management (CXM) encompasses strategies to consistently deliver high-quality experiences to customers across all touchpoints. Beyond simply improving satisfaction or usability,

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CXM aims to foster customer loyalty and drive sustained business growth [16, 18]. Advancements in digital services and technology enable companies to collect and analyze customer behavior and perception data in real time [28]. This allows businesses to derive actionable insights into customer interactions from the perspective of user experience and leverage them to optimize overall experiences [11].

However, significant challenges remain in CXM. It is difficult to holistically consider customer experiences across all touchpoints—from the initial brand awareness to purchase, usage, and post-service phases—and derive meaningful insights to enhance those experiences [19]. Furthermore, with the recent emphasis on Total Experience (TX), which integrates both customer experience (CX) and employee experience (EX) across various channels, experience management has become even more intricate. This integration reflects the connection between CX and EX, where a positive employee experience fosters greater employee engagement and improved service delivery, ultimately contributing to an enhanced customer experience [6, 8].

To address these challenges, this case study aims to develop a metrics framework to measure customer experiences in a company that offers diverse web-based services. Based on a literature review, we designed a metrics framework that addresses the subjects, phases, and aspects of experience for comprehensive CXM.

We conducted a user study with practitioners to evaluate the framework. Feedback from the practitioners was generally positive and offered valuable insights for applying the framework in practice. This feedback highlighted needs such as flexibility in adjusting metrics based on service contexts, effective data visualization, and enhanced data exploration capabilities. Finally, we discuss an AI-Enhanced Collaborative CXM model that integrates AI's analytical power with practitioners' domain expertise to generate actionable insights for improving customer experiences.

2 Background and Related Work

2.1 Bridging CXM and HCI

Customer Experience Management (CXM) focuses on analyzing the entire customer interaction process and improving experiences across touchpoints [11]. Similarly, Human-Computer Interaction (HCI) aims to understand how users interact with digital systems to enhance usability and overall interaction quality [14, 15, 31, 34]. Both fields share the goal of improving experiences through understanding human behavior and perception.

This shared purpose creates opportunities for mutual contributions. HCI provides CXM with established principles [35] and methodologies such as usability testing [21] and user journey mapping [13]. Conversely, the data and insights generated through

CXM serve as valuable resources for HCI research. CXM captures real-world customer interactions and feedback, providing data that can be used to evaluate the practical applicability of HCI research [28]. Such insights also help refine HCI principles and methodologies [33], and even suggest new research directions [25].

By integrating methods and insights, CXM and HCI collaboratively advance their approaches, fostering more practical and impactful strategies to improve human-centered experiences.

2.2 Considerations for Comprehensive CXM

Researchers have proposed various models to understand and measure customer experience. SERVQUAL is one of the well-known models, which evaluates service quality from the customer's perspective [23]. However, SERVQUAL considers services in physical environment, prompting the development of models for web-based [24] and mobile-based services [12]. Over time, the concept of customer experience has expanded beyond addressing service quality to include broader aspects such as emotions, perceptions, and expectations throughout the customer journey [2, 36]. Literature review studies have further synthesized these diverse aspects, proposing new dimensions or frameworks to achieve a more holistic understanding of customer experience [5, 30].

In the industry, efforts have also been made to classify and measure customer experience in practical terms. For example, Forrester Research, a leading market research firm, divides customer experience into three dimensions: Interaction, Perception, and Outcome. Interaction represents what takes place during the customer experience (e.g., website errors), perception reflects the customer's feelings about the experience (e.g., satisfaction levels), and outcome reflects the actions customers take as a result of their experience (e.g., churn rate). This classification provides businesses with a practical approach to analyze and enhance customer experience by addressing specific dimensions [7].

Additionally, the concept of Total Experience (TX) has recently gained prominence, emphasizing not only Customer Experience (CX) but also Employee Experience (EX) [8]. Positive employee experiences can directly lead to better customer service, making EX a crucial element of effective CXM [8]. Employee experience encompasses all interactions an employee has with the company, both direct and indirect, covering a wide range of aspects such as physical environment, access to technological tools, and cultural environment [9], and discussions around EX have continued to evolve as work culture and practices change due to digital transformation [3].

Customer journey is another critical aspect of CXM, encompassing all interactions from the customer's first touchpoint with the brand to purchase, usage, and post-service interactions. By mapping these stages, companies can analyze how customers engage with the brand and design more effective experiences [17]. While academic research often divides the customer journey into pre-purchase, purchase, and post-purchase stages [4], industry practices often adopt a more granular approach for enhanced management. For instance,

companies like Qualtrics¹ and Adobe² manage a detailed breakdown of the customer journey to optimize customer interactions at each stage.

2.3 CXM in Practice

In practice, companies have traditionally employed strategies like Customer Relationship Management (CRM) to manage customer data, improve customer relationships, and provide tailored services. As a sales-oriented approach, CRM has been instrumental in building stronger customer connections. However, with the growing emphasis on Customer Experience (CX), businesses have expanded their focus from merely managing relationships to enhancing overall customer experiences through Customer Experience Management (CXM) [29].

To achieve this, companies leverage customer feedback metrics such as Net Promoter Score (NPS), which measures customer loyalty, Customer Effort Score (CES), which evaluates the ease of customer interactions, and Voice of Customer (VOC), which captures direct customer feedback, as well as behavioral data like clickstream analytics, which tracks users' navigation patterns. While these metrics are effective for analyzing specific touchpoints, they often fall short in providing a holistic view of the entire customer journey. Isolated data points make it challenging to uncover the causes of customer issues and understand how experiences at different touchpoints collectively influence the overall experience [11, 19, 28].

To address these limitations and gain a more comprehensive understanding of customer experience, we propose a metrics framework that integrates the subjects of experience, phases of experience, and aspects of experience.

3 Development of Metric Framework for CXM

The company in this case study provides various web-based SaaS (Software as a Service) solutions, including logistics, cloud, and work process automation (RPA). These solutions come with a wide range of features and complexities. A major challenge was to create a metric framework that could capture diverse customer experiences, considering both the specific context of each service and an End-to-End (E2E) perspective, which examines the entire customer journey from start to finish. This section details the process of developing the framework.

3.1 Process of Metric Framework Development for CXM

3.1.1 Step 1: Literature Review and Analysis. We began by conducting a literature review using major academic databases such as Google Scholar, ACM Digital Library, and Web of Science, focusing on keywords like "customer experience," "employee experience," and "customer journey." Additionally, we reviewed industrial reports from consulting firms such as Forrester and Gartner to understand recent trends and practical approaches in CXM. These sources provided both theoretical foundations and real-world insights for developing the metric framework.

¹<https://www.qualtrics.com/experience-management/customer/customer-journey-stages>

²<https://business.adobe.com/blog/basics/customer-journey>

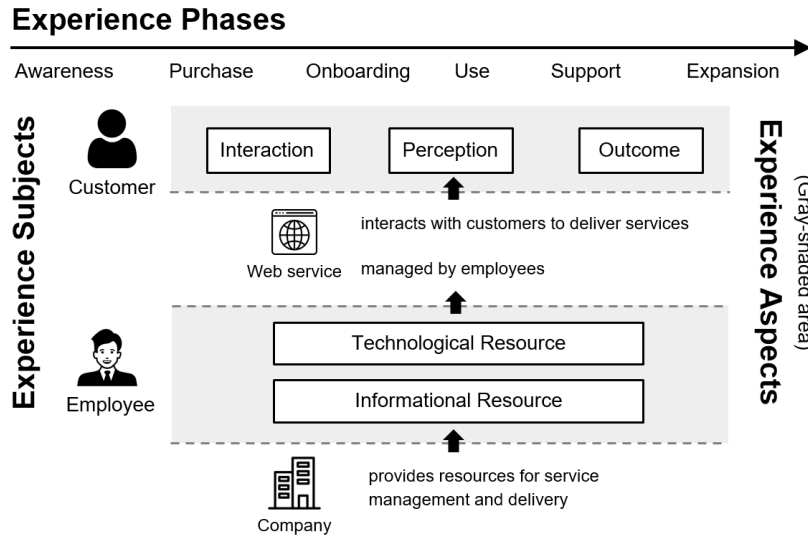


Figure 1: A view of experience subjects, phases, and aspects in service delivery.

3.1.2 Step 2: Structuring the Main dimensions. We structured the framework with three main dimensions: (1) subjects of experience, (2) phases of experience, and (3) aspects of experience. The subjects of experience defines who is experiencing, encompassing both customers who directly interact with the service and employees who manage the service. The phases of experience captures when experiences occur, aligning with the six stages of the customer journey managed by the case company: Awareness, Purchase, Onboarding, Use, Support, and Expansion. Awareness represents the initial phase where customers become aware of the brand, while Purchase involves acquiring services. Onboarding is critical for this company due to the complexity of its solutions, requiring customers to familiarize themselves with various features. Use refers to ongoing interactions with the service, Support addresses issues or problems that arise, and Expansion represents the phase where customers make additional purchases or extend their existing services. Finally, the aspects of experience addresses what type of experience is occurring, focusing on what happens during interactions, how customers perceive these interactions, and what actions they take as a result, categorized into Interaction, Perception, and Outcome [7, 30].

For Employee Experience (EX), we focused on aspects specifically related to service delivery and management. Since EX encompasses a broad range of factors—such as relationships with colleagues, technology use, and physical work environments [9]—addressing all of these was impractical. Instead, we concentrated on two dimensions: informational resource and technological resource. Informational resource refer to the guidance and knowledge employees receive to effectively support the customer journey, while technical resource encompass the tools and technologies provided to enable efficient service management. Figure 1 illustrates the overall structure of the framework, highlighting the interplay between CX and EX dimensions during the service delivery process.

3.1.3 Step 3: Identifying and Selecting CX Constructs. Based on our literature review, we identified numerous CX constructs and condensed them into 37 unique constructs through thematic analysis. From these, we selected 12 constructs that aligned with the case company’s context. We excluded constructs that were overly abstract or broad (e.g., service experience) due to their lack of specificity, as well as those that were difficult to measure (e.g., personal factors) or not directly relevant to web services (e.g., physical structure).

3.1.4 Step 4: Mapping CX Constructs to Main Dimensions. Next, we mapped the constructs selected in Step 3 to the main dimensions defined in Step 2. For example, the Interaction dimension includes aspects of user actions and service usability. Accordingly, Efficiency, which measures how quickly users can navigate a web page, was assigned to Interaction. This mapping process offered a more detailed understanding of customer experiences by connecting each main dimension to specific constructs. The 12 selected constructs and their corresponding dimensions are shown in Table 1.

3.1.5 Step 5: Designing Metrics Based on Service Contexts. Based on the selected constructs, we designed metrics tailored to the entire customer journey. We reviewed existing measurement scales for each construct. For example, Efficiency was measured using items such as ‘This site makes it easy to find what I need’ [24]. These scales can be measured using real-world data, such as clickstream data and survey responses, to quantify the constructs.

Given the complexity of customer experience, single metrics often could not capture all nuances. Thus, we defined composite experience metrics by combining several underlying components. For example, in the Awareness phase, we designed the *Content Navigation Efficiency*, which combines metrics like *search result adoption rate*, *content search time*, and *recommended content adoption rate*. This metric can be measured using behavioral data such as clickstream data, which captures how users interact with and navigate

Table 1: CX constructs for CXM in web-based services

Aspect	Construct	Focus of Measurement
Interaction	Service availability	How technically reliable is the service provided to customers?
	Efficiency	How easily and quickly can customers complete their desired tasks using the service?
	Fulfillment	How well does the service deliver on promised features or outcomes and provide benefits to customers?
	Behavioral intention	How willing are customers to engage in intended behaviors (e.g., visiting pages, viewing content) as prompted by the service provider?
	Employee	How do customers evaluate the competence and attitude of employees (e.g., professionalism, friendliness)?
	Contact	How easily can customers access support when they need help?
	Responsiveness	How quickly and effectively does the service address issues experienced by customers?
Perception	Satisfaction	How satisfied are customers with their overall experience interacting with the service?
	Confidence	How much do customers trust the service and feel confident that it will meet their needs?
Outcome	Word of mouth	What attitudes or emotions are reflected in customers' comments or opinions shared after their service experience?
	Loyalty intention	How willing are customers to repurchase the brand or service in the future?
	Recommendation intention	How willing are customers to recommend the service to others?

through content. Additionally, for metrics like *Content Usefulness* that are difficult to measure through behavioral data, we adopted survey data. For metrics that are challenging to quantify, such as *Customer Sentiment*, we considered using sentiment analysis.

Table 2 provides examples of the metrics used across different phases of the customer journey. A full list of metrics covering the entire journey is presented in the Appendix.

4 Evaluation of The Experience Framework

We performed a user study with practitioners to evaluate the developed metric framework. A total of ten participants were recruited through snowball sampling, including four women and six men. The participants represented diverse departments, with five from logistics services, two from cloud services, one from automation solutions, and two from customer experience management. Their professional roles also varied, encompassing marketing (four participants), planning (three participants), customer experience management (two participants), and development (one participant). Among these, three participants held managerial positions.

We conducted semi-structured interviews, with each session lasting around 40 minutes. We focused on (1) how easily participants understood the experience metrics, (2) their perceptions on the appropriateness of the metrics, and (3) considerations when applying the metrics in practice. Interview data were transcribed using an automatic transcription tool, followed by manual corrections for accuracy. The affinity diagramming method was used to categorize responses and identify key patterns and themes.

4.1 Main Results

4.1.1 Overall Perception on The Experience Framework. We presented the structure of the framework to participants and asked participants whether they easily understood this structure, and all of them reported that it was easy to comprehend. Regarding the appropriateness of the metrics, 9 out of 10 participants found them

appropriate. The remaining participant (P2), who worked in development, found it difficult to evaluate the metrics' appropriateness due to the technical focus of his role. One marketing practitioner noted, "*The underlying components that make up the main experience metrics seem well-chosen. For instance, just monitoring the user acquisition index could help us assess whether the service is performing well*" (P6). Additionally, one participant in a managerial role mentioned, "*When interviewing or discussing with staffs, these kinds of factors are frequently brought up*" (P7), referring to the general relevance of the employee experience metrics.

We presented the structure of the framework to participants and asked if they found it easy to understand. All participants reported that the structure of metrics was easy to comprehend. Regarding the appropriateness of the metrics, nine out of ten participants considered them suitable. The remaining participant (P2), who worked in development, found it challenging to evaluate the metrics due to the technical focus of his role. One marketing practitioner commented, "*The underlying components that make up the main experience metrics seem well-chosen. For instance, just monitoring the user acquisition index could help us assess whether the service is performing well*" (P6). Additionally, one participant in a managerial role mentioned, "*When interviewing or discussing with staffs, these kinds of factors are frequently brought up*" (P7), referring to the general relevance of the employee experience metrics.

4.1.2 Adjusting Experience Metrics Based on Service Contexts. Participants generally provided positive feedback on the experience metrics and their underlying components but emphasized the need for adjustments based on service characteristics, maturity level, and their specific roles. Participants mainly reported three opinions for metric adjustment: (1) certain underlying components may be less relevant for specific services, (2) additional components may be needed for certain contexts, and (3) the weighting of components should be adjusted according to the maturity of the service.

Table 2: Examples of the CX and EX metrics used across different phases of the customer journey

Subject	Phase	Metric Name	Definition	Underlying Components
Customer	Awareness	Content navigation efficiency	The degree to which customers can efficiently navigate through content	Average search result adoption rate, Average content search time, Recommended content adoption rate
		Content usefulness	The extent to which content is perceived as beneficial and helpful to customers	Degree of content usefulness (survey)
		User acquisition	The effectiveness of attracting visitors to the web service	Number of visitors, Organic acquisition rate, Bounce rate
		Customer sentiment	Evaluation of customer sentiment regarding their experience during the awareness phase	Sentiment analysis results
	Purchase	Purchase process optimization	Efficiency of the purchasing process for customers	Average purchase process completion rate, Purchase process dropout rate, Time to complete purchase
		Purchase intention	Level of customer actions/tasks related to purchasing	Number of quote requests, Number of purchase requests, Number of free trial sign-ups
	Onboarding	Onboarding engagement	Level of customer engagement with onboarding content	Number of participants in onboarding content, Onboarding content completion rate, Average progress rate of onboarding content
		Customer confidence	Confidence level in using the service after the onboarding phase	Degree of increased confidence after engaging onboarding content (survey)
	Use	Function usage efficiency	The degree to which customers efficiently use service functions	Average time to complete a task, Task success rate, Average number of clicks to complete a task
		Customer loyalty	The customer's intent for continued use of the service	Repurchase rate, Retention rate over a specific period
	Support	Automated support efficiency	The efficiency of automated support methods, such as chatbots, in resolving customer issues	Average time to resolve issues, Issue resolution success rate, Rate of repeated requests for manual support
		Support responsiveness	Measures the speed and effectiveness of responses to customer support requests	Average time to resolve issues after request submission, First-attempt issue resolution rate
	Expansion	Extended purchase conversion	The conversion rate from customer interest in additional features or services to actual purchase	Rate of interest in additional features, Conversion rate of additional feature purchases
		Net promoter score (NPS)	The likelihood of recommending the service to others based on the expansion experience	Degree of likelihood to recommend the service (survey)
Employee	Can be utilized across all phases	Informational resource accessibility	The extent to which employees can easily access work-related knowledge assets to support their work	Degree of ease in assessing work-related knowledge assets to support work (survey)
		Informational support sufficiency	The extent to which employees are adequately provided with work-related assets	Degree of adequacy in the provision of work-related knowledge assets (survey)
		Informational resource utilization	The extent to which work-related knowledge assets contributes to improving employees' job productivity	Degree of contribution of work-related knowledge assets to improving productivity (survey)
		Technical resource accessibility	The extent to which employees can easily access and utilize technical tools for their work	Degree of ease in accessing technical tools (survey)
		Technical support sufficiency	The extent to which employees are adequately and sufficiently supported in utilizing technical tools	Degree of support provided for utilizing technical tools (survey)
		Technical resource utilization	The extent to which technical tools contributes to improving employees' job productivity	Degree of impact technical tools have on improving employees' job productivity (survey)

Some participants acknowledged the overall importance of experience metrics but noted that specific underlying components might hold less relevance for their services. For example, one marketing manager commented, *"User Acquisition is definitely needed, but bounce rate isn't as significant for our service"* (P5). Others highlighted the need for additional role-specific component. One planning practitioner, for instance, stressed the importance of *Automated Support Efficiency* during the Support phase, stating, *"It's crucial to see how well our chatbot is answering. For example, it might give irrelevant answers, so we monitor how accurate its responses are"* (P7). Participants also emphasized the need to adjust weightings of underlying components based on the service maturity. One marketing manager explained, *"The importance of each component changes depending on the service stage. For example, for User Acquisition, visitor numbers are crucial in the early stage, but as the service matures, other components become more significant, so the weightings should change"* (P3). These results suggest that while the framework is well-structured with relevant experience metrics, it needs to remain adaptable, allowing for the inclusion, modification, or adjustment of components weightings based on the unique context and stage of each service.

4.1.3 Need for Effective Data Presentation. We identified three key needs for effectively applying the framework: (1) clear explanations of each metric's meaning, (2) intuitive data visualization methods, and (3) the capability to explore detailed information.

First, participants emphasized the need to understand what each metric reflects about the customer experience. One participant remarked, *"The experience metrics sound convincing, but the concept of 'Function Usage Efficiency' can be a bit vague. For instance, efficiency could refer to task completion time or success rates"* (P4). This feedback highlights the importance of providing clear explanations so that practitioners can quickly interpret metric changes and make informed decisions. Second, participants noted that simple numerical data is often insufficient for understanding trends or changes in certain situations. They emphasized the need for visual presentations that clearly convey patterns. Four visualization methods were suggested: time series graphs for tracking changes over time, comparisons across different periods to identify trends, comparisons with other services within the company to evaluate relative performance, and visual displays of progress toward specific goals to help assess target achievement. Lastly, participants highlighted the need for drill-down capabilities to access fine-grained data. While composite experience metrics offer a general overview, participants found it challenging to delve into detailed information. One CX manager explained, *"The concept of experience metrics is good, but without a way to drill down and see individual cases or situations behind the numbers, it's hard to fully utilize them"* (P9). This highlights the need for tools that allow practitioners to explore the detailed components of each metric.

4.1.4 Need for AI-based Technology Support. Interestingly, participants expressed a need for AI assistance in utilizing the framework, especially for (1) analyzing the causes behind changes in metrics and (2) providing actionable recommendations for experience improvement.

Regarding cause analysis, participants wanted AI to comprehensively analyze the underlying components that contribute to

changes in experience metrics. One participant mentioned, *"It would be helpful if the metrics could also explain which aspects were high or low to result in the given value"* (P1). Some participants wanted a deeper understanding of the factors behind changes in metric values. One participant explained, *"It would be really helpful if AI could summarize new features released or customer feedback during the periods when metrics change. This way, I could gain insights without having to analyze everything manually"* (P9). This highlights the potential for AI to automatically analyze and provide context behind metric changes, enabling practitioners to derive insights more efficiently. Participants also emphasized the value of AI-generated recommendations for improving metrics. One participant suggested, *"When a metric is reported, it would be helpful if the AI could also suggest what actions we could take to improve it"* (P6). Despite being aware of AI's limitations, they saw value even in general suggestions. As one noted, *"If it's difficult to consider specific service contexts, even general guidance would be helpful. I could then adapt it to our specific service contexts"* (P1). This indicates that even preliminary AI-generated recommendations can provide a useful starting point, allowing practitioners to further refine insights for their unique service contexts.

5 Discussion

5.1 A Comprehensive Approach to Experience Management

This case study introduces a metrics framework designed to offer a comprehensive approach to managing customer experiences. The framework evaluates experiences through three key dimensions: (1) subjects of experience, (2) phases of experience, and (3) aspects of experience.

The subject includes both customers and employees, integrating Customer Experience (CX) and Employee Experience (EX). By addressing both perspectives, the framework enables companies to understand how CX and EX interact, facilitating strategies to improve both. For instance, positive employee experiences can directly enhance customer experiences, providing a more holistic view than traditional customer-only approaches. The phase dimension organizes experiences across the stages of the customer journey, allowing companies to track changes over time and pinpoint issues specific to each stage. This structure makes it easier to identify areas for improvement and apply targeted strategies, offering a clearer understanding of how experiences evolve throughout the journey. The aspect dimension distinguishes interaction, perception, and outcome, providing insight into how customers interact with services, their feelings about those interactions, and the resulting behaviors. This categorization enables deeper insights by taking into account both the internal responses of customers and the actions they take. For example, analyzing usability data alongside satisfaction ratings can help pinpoint the causes of customer dissatisfaction more accurately.

5.2 Challenges for Practical Application of The Framework

While practitioners provided positive feedback on the framework's appropriateness and usefulness, they also identified several key challenges that must be addressed for its effective application.

First, meaningful presentation of data is essential to derive actionable insights from the metrics. Practitioners emphasized the importance of clear explanations about what each metric represents in terms of customer experience. They also highlighted the need for intuitive visualizations that allow for quick understanding of key insights. Suggested visualization methods included time series graphs, comparisons between services, and progress tracking toward goals. These findings indicate that the visualization of experience metrics should be tailored to the specific intentions and purposes of practitioners [1].

Second, practitioners pointed to the need for drill-down capabilities to explore the detailed information behind the metrics. Such functionality enables a deeper understanding of the causes behind changes in metrics and specific situations, leading to more targeted insights. This requires interactive data visualization tools, which can be developed by referring to practical data visualization design principles [27]. Moreover, improving data literacy among practitioners through company-level support is crucial to maximize the effectiveness of the framework [10].

Third, flexibility in adjusting the metrics is necessary to accommodate the characteristics and maturity levels of different services. While participants generally agreed with the overall structure of the metrics, they emphasized that specific compositions or weightings of underlying components should be adaptable to the unique context of each service. This suggests that the framework should maintain its overall structure while allowing for adjustment in its application. For more detailed analyses of customer experience, methods such as feature engineering and integration with domain-specific knowledge should also be considered to ensure the metrics remain relevant and accurate [20].

5.3 Envisioning AI-Enhanced Collaborative CXM

In our user study, practitioners expressed the need for AI support in analyzing experience metrics and providing actionable recommendations. They emphasized that even general analyses and AI guidance could be valuable when combined with their domain expertise, resulting in more accurate and context-specific insights.

These findings suggest possibilities for the future of how human workers engage with AI in the workplace. This aligns with existing research on human-AI collaboration, which argues that the future of AI in knowledge work should not focus solely on full automation. Instead, it should emphasize collaborative approaches where humans and AI work closely together. For instance, one study highlighted that enhanced human-AI collaboration leads to increased productivity, demonstrating the potential of such partnerships [26]. Similarly, this perspective aligns with the principles of *Arbeit 4.0*, a concept that explores the evolving relationship between humans and machines in the workplace [22]. *Arbeit 4.0* emphasizes optimizing collaboration by ensuring that employees are equipped with

the *experiential knowledge* needed to work effectively alongside intelligent systems.

In the context of Customer Experience Management (CXM), as CXM expands from its origins in CRM to encompass more holistic approaches like TX-based CXM, there is potential for an AI-Enhanced Collaborative CXM model. This model envisions AI providing preliminary analyses and insights, while practitioners refine and adapt these insights based on their understanding of specific contexts. This collaborative process allows for a deeper understanding of customer needs and more effective strategies to improve experiences. The envisioned approach is illustrated in Figure 2. For this collaboration to be effective, clear communication interfaces between AI systems and practitioners are crucial, as is transparency in how AI-generated insights are produced. These factors help to ensure that practitioners can understand, trust, and act on the insights provided by AI [32]. In summary, while this study suggests that an AI-Enhanced Collaborative CXM model holds promise, the success of such an approach will depend on thoughtful integration of AI's capabilities with the nuanced understanding that practitioners bring to their work.

5.4 Reflection and Limitations

In this study, we aimed to design metrics for CXM based on constructs defined in the academic field of customer experience. One of the most significant challenges was the gap in granularity between the constructs proposed in academic research and the practical metrics needed for real-world applications. Academic constructs are typically general in nature, whereas practitioners require fine-grained metrics to address specific business goals. However, focusing too narrowly on specific problems during metric development may limit a comprehensive approach to customer experience. Our goal was to create a framework that balances the generality of academic constructs and the specificity required in practice, enabling comprehensive customer experience management. Achieving this balance was challenging, as it was difficult to find existing studies that effectively bridge these two areas.

The key to bridging theory and practice lies in domain knowledge. However, domain knowledge is often embedded in the personal experience of practitioners rather than being systematically documented. Additionally, conducting in-depth interviews to extract this knowledge requires significant human and time resources, which were limited in our case. To overcome these challenges, we reviewed and organized records from CXM-related activities, decision-making processes, and communications with practitioners. Through a thorough analysis of these records, we built a foundation of domain knowledge. We then integrated this domain knowledge into the theoretical framework established through our literature review to design the metrics. When gaps in knowledge arose, we addressed them by directly consulting practitioners and asking targeted questions. This process allowed us to effectively bridge the gap between theory and practice.

Bridging theory and practice provided a distinct advantage. By incorporating domain knowledge into the metrics framework, practitioners found the framework to be highly relevant and appropriate

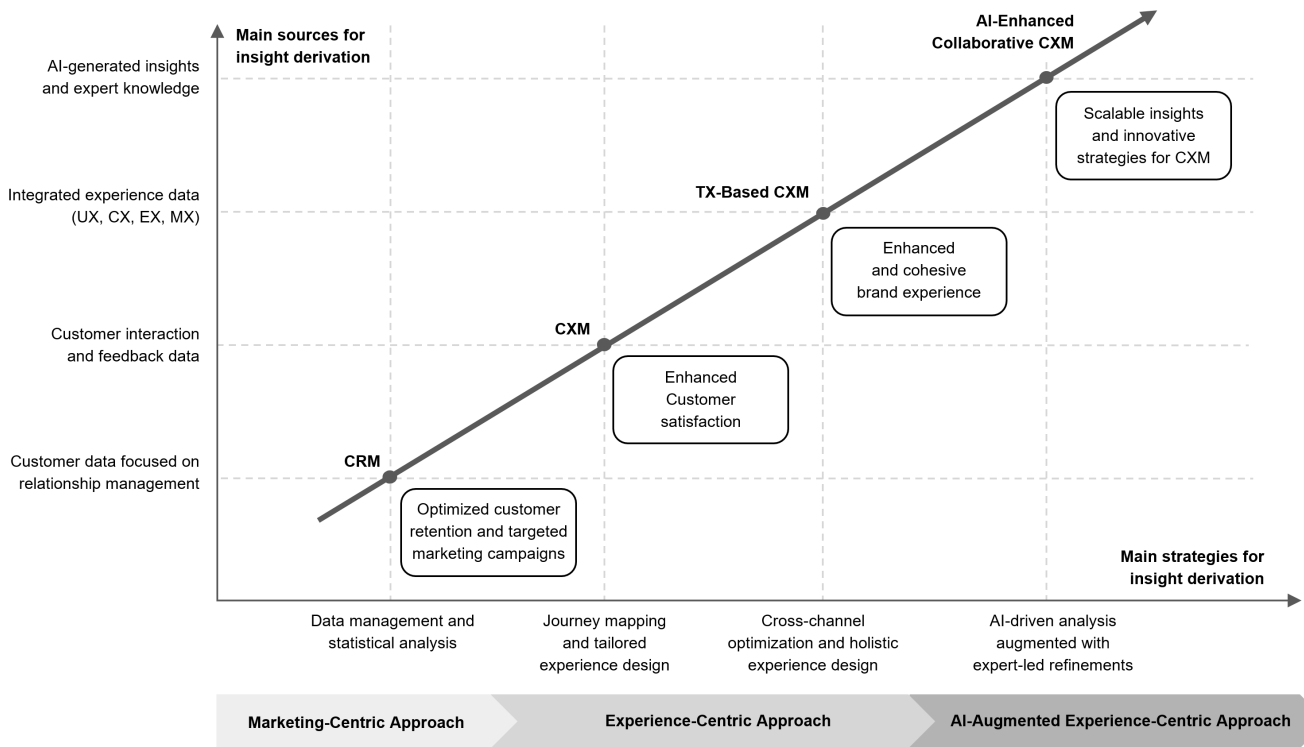


Figure 2: Advancing approaches to customer experience management through evolving sources and strategies for insight derivation

for their needs. We believe this case study demonstrates how combining theoretical and practical perspectives can result in a robust and adaptable approach to customer experience management.

The proposed metrics framework was developed with a focus on web-based services, which limits its direct applicability to other types of services. However, because the framework integrates academic theory with practical domain knowledge, it has the potential to be extended to other service types. The framework development process involved two key tasks: (1) establishing a theoretical foundation for CXM and (2) designing metrics that reflect domain knowledge and service contexts. While the domain-specific focus of the metrics may hinder their immediate applicability to other services, the framework's general theoretical foundation allows room for adaptation.

Most services share commonalities, such as customers who use the service, employees who operate it, and the existence of a customer journey. During the journey, interactions, perceptions, and outcomes can typically be observed. These shared elements form the foundation of the framework's main dimensions—subjects of experience, phases of experience, and aspects of experience. If metrics were designed solely based on domain-specific knowledge, applying them to other service types would require starting from scratch. However, our framework provides a universal foundation that can be adapted to diverse contexts by incorporating additional dimensions as needed. For example, in services that combine online and offline experiences, the core dimensions remain the same, but

an additional dimension, such as "experience medium," could be introduced to capture where the service is experienced. This flexible structure allows detailed metrics to be tailored to the unique context of each service. By repeating the five steps of our framework development process, we believe the framework can be extended and adapted to accommodate other service types.

6 Conclusion

This study proposes a framework that integrates subjects of experience, phases of experience, and aspects of experience, and evaluated its practical applicability through a user study with practitioners. We also identified several challenges that need to be addressed for real-world implementation. From a business perspective, this case study enables companies to implement systematic and comprehensive CXM, offering the potential to enhance customer loyalty and improve business performance. For the field of HCI, this study contributes by identifying research challenges, such as user data-based analysis and visualization, as well as human-AI collaboration. These challenges could become central topics of discussion in the practical application of CXM, offering valuable insights to the HCI research community. In the future, we plan to address these challenges by implementing a CXM system based on the proposed framework and applying it to case services. This will allow us to empirically verify the positive impact of comprehensive experience management on business performance. Additionally, we will explore the potential of

AI-Enhanced Collaborative CXM, combining AI's analytical capabilities with the domain expertise of practitioners, and gradually work towards developing this advanced approach to customer experience management.

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